

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in stochastic processes, time-dependent modeling, scalable likelihood-based inference, simulation, machine learning, and statistical diagnostics. Builds reproducible Python/PyTorch research pipelines and has hands-on experience with market making, Black-Scholes valuation, implied-volatility estimation, and inventory-risk controls.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Jan 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, reusable and restartable pipelines

Quantitative Methods: Stochastic processes, time-series and spatiotemporal modeling, Gaussian processes, MLE/GLS, Monte Carlo simulation, numerical optimization, regression and machine learning, spectral analysis, model diagnostics and validation

Options and Trading: Black-Scholes valuation, implied volatility, put-call parity, fair-value estimation, market making, inventory-aware quoting, and liquidation controls

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test framework for binary risk indicators, translating a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and statistically controlled stopping rules.
- Applied Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Analyzed model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

QUANTITATIVE RESEARCH AND TRADING

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy combining market-based fair-value estimation with inventory-aware quoting and rolling-window soft/hard liquidation controls.
- Implemented Black-Scholes call valuation and bisection-based implied-volatility estimation for option instruments across multiple strikes, and evaluated no-arbitrage relationships through put-call parity.

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Diagnostics

Sep 2024 - Present

- **Stochastic Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- **Simulation and Diagnostics:** Designed scale- and frequency-resolved tools to identify whether model misspecification arises from low-frequency structure, high-frequency behavior, noise, or dependence.
- **Data and Research Engineering:** Built Python/PyTorch pipelines for quality filtering, missingness tracking, time-dependent alignment, estimation, and HPC execution; integrated compiled C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Probability Theory, Stochastic Processes, Machine Learning, Linear Algebra, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)